WOLF - Wolfspeed | Full investment artifact

Full investment research | Model assumptions and source evidence

Research date: 11 October 2026 capital review; engineering review 6 October; operating calibration 18 September | **Framework:** 8.0 | **Format:** standalone current-state analysis **Status:** Full current quarterly model; October 11 capital/share update; utilization and operating forecasts retained | **Classification:** SILICON CARBIDE TURNAROUND - high-voltage AI opportunity **Artifact Base:** 35/60 = **58% - Fail** | **Portfolio Excel Base:** 58% (current master; prior 55 override removed on October 7) **Snapshot:** supplied-workbook price USD 32.870; model share proxy 111.000000M; implied snapshot market capitalization 3,648.570 USD millions. Price is frozen from the supplied workbook, not a current-market refresh; capitalization uses the modeled denominator. **Active price target:** Q4 2027 **USD -22.602623**, pricing Q4 2027-Q3 2028; updated only for the conditional capital share proxy. All 15 quarterly price targets are derived below.

AI power demand is becoming more tangible through qualified customers. The improved opportunity is offset by a much larger potential share count and the distance between gross-margin breakeven and operating profitability.

October 11 update: conditional financing and the common-share claim

Wolfspeed's October 7 SEC filing describes a conditional 30-year facility of up to **USD 1.5B**. The initial **USD 600M** chiefly refinances first-lien 2030 notes and fees. The remaining **USD 900M** supports the project through three tranches of USD 200-400M. A commitment is not a completed loan draw. The issuer announcement and filing are the factual sources; the execution weights and dates below are analytical assumptions.

The facility requires **USD 750M** of qualifying contribution. Of that, USD 150M must be qualifying equity proceeds, with USD 50M before effectiveness and USD 100M before the second tranche. The other USD 600M has more flexible permitted sources, subject to OSC acceptance, including qualifying conversion face value or accepted excess cash. Qualifying sources must be received after June 28. The checked post-June primary filings do not establish a completed equity receipt satisfying the USD 50M or USD 150M thresholds. The September resale registration is not issuer equity proceeds.

Three joint capital outcomes, one retained share-proxy method

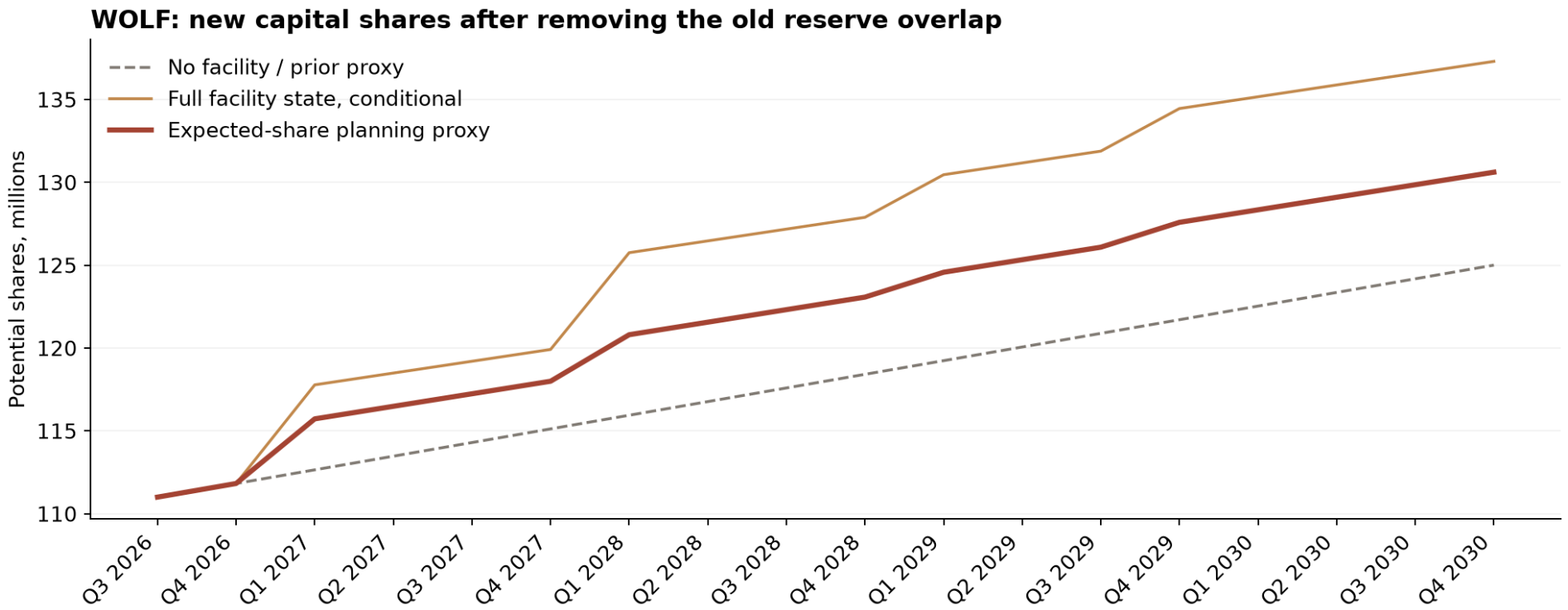
Outcome	Probability	Loan assumption	New contribution equity assumption
No facility closes	40%	No draw; original share proxy retained	None from this branch
Initial tranche only	20%	USD 600M in Q1 2027	USD 50M in Q1 2027
Full facility	40%	USD 600M in Q1 2027, then USD 300M in Q1 2028, Q1 2029 and Q4 2029	USD 50M in Q1 2027 plus USD 100M in Q1 2028

The 60% combined probability of initial funding recognizes the conditional government commitment and refinancing use, while preserving a material chance of failure. The 40% full-facility outcome also requires later project, contribution and documentary conditions. These are joint states, not independent probabilities applied twice. The draw dates are assumptions within the 36-month commitment period, not issuer guidance. Three USD 300M later tranches sit within the disclosed USD 200-400M bounds.

New contribution equity uses **USD 30.58 per share**, the October 9 regular-session quote, only as an issue-price proxy. The workbook's frozen USD 32.87 quote remains unchanged. The sensitivity is USD 24.464-36.696, or 20% either side. No completed raise at these prices is claimed. Conditional issuance is 1.635056M shares for USD 50M and 4.905167M for USD 150M, before fees. Fees, exact preferred/common form and final priced terms remain unresolved.

The government warrants total **7.5% of fully diluted shares including those warrants**, excluding common issued to satisfy the Minimum Contribution. Therefore W/(B+W)=7.5%, or W=B×7.5%/92.5%. The illustrative grant base is the existing 110.860M gross converted bridge plus 3.766M outstanding awards, or **114.626M**. It excludes 8.332M ungranted future award capacity and all new contribution shares. The result is **9.294M nominal full-facility warrants**, issued pro rata with funded loan tranches. Actual effectiveness-date securities and final definitions must replace this June proxy. The strike is a mutually agreed VWAP not yet fixed; no exercise cash is assumed, and cashless settlement can differ from nominal warrant shares.

The old 111M-to-125M share schedule already reserved 12.098M for outstanding and future awards, leaving **1.902M** for capital. The old capital reserve grows linearly as 1.902M × i/17, where i=0 in Q3 2026. Each funded state adds only MAX(0, new contribution shares + pro-rata warrants - old capital reserve). The no-facility state retains the full old denominator. The active share proxy adds 20% of the initial-state net increment and 40% of the full-state net increment. It is an **expected-share planning proxy**, preserving the existing convention; it is not a probability-weighted average of state valuations. Inverting expected shares and averaging individual share values are mathematically different.



WOLF capital share bridge. Conditional states are model assumptions; nominal potential shares are not actual shares issued.

Quarter	Prior share proxy M	Initial-state capital shares M	Full-state capital shares M	Old capital reserve M	Active share proxy M
Q3 2026	111.000000	0.000000	0.000000	0.000000	111.000000
Q1 2027	112.647059	5.352656	5.352656	0.223765	115.724393
Q4 2027	115.117647	5.352656	5.352656	0.559412	117.993593
Q1 2028	115.941176	5.352656	10.481567	0.671294	120.801558
Q1 2029	119.235294	5.352656	12.340367	1.118824	124.570678
Q4 2029	121.705882	5.352656	14.199167	1.454471	127.583398
Q4 2030	125.000000	5.352656	14.199167	1.902000	130.608998

Operating economics and the remaining funding condition

Revenue, product gross profit, ordinary operating expenses, the **15x GAAP-style F4Q OP** method and both Base scores remain unchanged. Refinancing proceeds and interest savings are not operating revenue or OP. The USD 900M project portion has no disclosed accepted-unit, budget or customer-delivery bridge sufficient to add another operating ramp above the existing AI and materials forecasts. The current AI ramp already requires capital; a financing announcement is not a second demand shock. This review assigns no additional project revenue, gross profit or operating expense.

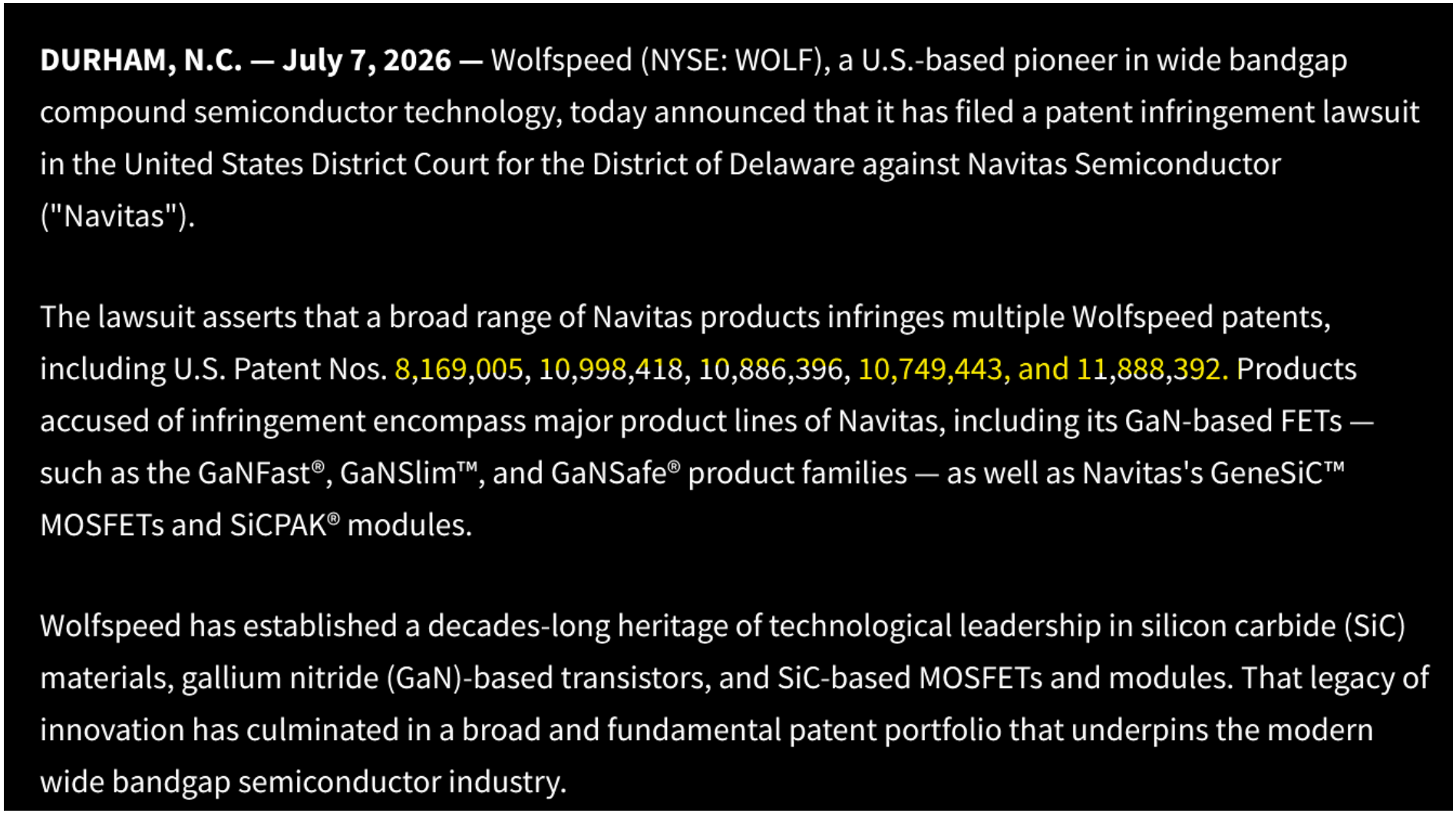
In the full state, the remaining **USD 600M qualifying contribution** is assumed to be satisfied by already-included conversions or eligible excess cash accepted by OSC. This eligibility is unresolved. If that source is rejected or insufficient, further equity, different financing or delayed project execution would be required. Existing conversion reserves are not added twice. This share exercise does not certify liquidity, covenant headroom or project funding availability. Final agreement, effective-date fully diluted securities, warrant strike, actual equity price/fees and contribution acceptance are the precise next evidence needs.

The active Q4 2027 F4Q OP remains **minus USD 177.797647M**. Its share proxy rises from **115.117647M to 117.993593M**, moving the formula result from **minus USD 23.167297 to minus USD 22.602623**. That movement toward zero is a denominator effect, **not an operating improvement or positive investment upside**. For the positive Q1 2030 window, unchanged OP of USD 545.651765M produces **USD 63.774267**, down from USD 66.798464, a **-4.53% dilution effect**.

Current Excel Base is **58**, matching the saved master after the owner removed the prior 55 override on October 7. Analytical Base is **58**. Correcting obsolete artifact references to 55 does not change either score.

October 6 evidence: the Navitas complaint and existing cost allowance

Wolfspeed’s July 7 complaint asserts **five U.S. patents**, not the six described in the August researcher discussion. The issuer identifies patents 8,169,005; 10,998,418; 10,886,396; 10,749,443; and 11,888,392. The dispute concerns specified GaN and SiC products. A patent abstract, an approaching expiry date or the claimant’s current product mix does not establish claim validity, infringement, damages or an injunction. The article’s categorical legal dismissal is therefore not a financial-model input.



Wolfspeed complaint announcement lists five patents; allegations are not a court finding

Wolfspeed complaint announcement lists five patents; allegations are not a court finding Source: Irrational Analysis, August 9, 2026, original exhibit 9. The saved image is reproduced without modification.

The Wolfspeed fiscal2026 filing describes the case and records no potential recovery. The Navitas Q2 filing disputes the allegations, identifies the affected product families and says a reasonably possible loss cannot yet be estimated. Its separate Renesas lawsuit is not this case. These are opposing parties' disclosures, not an adjudicated outcome. No royalty rate, recurring licensing contract or quantified damages award is established.

Cost and overlap treatment. The lawsuit is an observed event, recorded at **100% event status**, not a 100% probability of success. The saved model uses August 19 operating-expense guidance, after the July 7 complaint, and the filing explicitly includes litigation/outside-counsel spending in SG&A. The model begins at $72M quarterly opex, including its stated recurring GAAP and restructuring allowances, and rises over time. The article supplies no new litigation event or identified cost beyond that aggregate allowance.

For scale, a hypothetical **$1-3M annual legal-cost slice** would represent **0.31-0.94%** of the existing **$319.294M CY2027 operating-cost forecast**; $2M would be about 0.63%. This is a diagnostic of the possible composition of current SG&A. It is not a reported legal budget, a new forecast branch, an unused probability-weighted expense or a new charge added to the model.

Treatment	Revenue change	COGS change	Opex change	OP change
Same July complaint reviewed against later cost anchor	0	0	0	0
Incremental royalty or damages windfall	0	0	0	0

All changes in the table are USD millions. Zero reflects overlap with the later aggregate cost anchor and the absence of a new economic event, rather than a claim that litigation is free. An identifiable increase in legal spending beyond guidance would enter opex; any eventual non-recurring recovery would require separate treatment and would not automatically be capitalized as recurring earnings at 15x. The current model does not forecast a legal victory or immunity from adverse claims.

The October 6 operating review added no new litigation loss. The October 11 capital update retains forward OP of minus USD 177.797647M and the 15x method, while increasing the starting-quarter share proxy to 117.993593M. The active Q4 2027 result is USD-22.602623 per share. A negative result supplies no positive earnings-based support; its movement toward zero after dilution is not an earnings upgrade. Analytical Base 58 and current Excel Base 58 remain unchanged.

The next useful evidence is case progress that changes the economic exposure, actual litigation spending against the current cost anchor, a defined recurring license, or a supplier/customer shift tied to a qualified power device. General GaN-versus-SiC rhetoric does not replace the existing application-specific sales and utilization model.

October 6 engineering review: what a 10 kV device can earn

The April 25 and May 17 Irrational Analysis articles identify a real opening for high-voltage power semiconductors, but several calculations do not support their supplier rankings. The full texts and all 103 associated body images were reviewed, including datasheet figures at original resolution where their units or operating conditions mattered. Datasheets are component evidence; a qualified converter, its delivered cost and the customer's orders determine Wolfspeed's earnings. [P1, P2]

Three distinctions change the analysis. Maximum transistor power dissipation measures permissible heat loss under specified cooling conditions, not the useful electrical power delivered by a converter. Single-pulse safe-operating-area curves marked D=0 do not establish a continuous switching-frequency range by taking the reciprocal of pulse duration. The April comparison also copies output charge, Qoss of 45/47 nC, into a row labelled capacitance, Cds, then uses a capacitance-loss formula. Charge and capacitance have different units; that graph cannot rank commercial GaN and SiC efficiency. Infineon's GaN datasheet specifies Coss, Qoss and Eoss separately, and its application note defines the single-pulse boundary. [P3, P4]

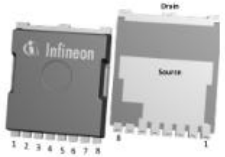
Public
IGT65R045D2
Final datasheet

CoolGaN™ G5

CoolGaN™ Transistor 650 V G5

Infineon's CoolGaN™ is a highly efficient gallium nitride (GaN) transistor designed for power conversion at 650 V. It enables higher power density, supports reduced system BOM cost, and facilitates miniaturized form factors. Produced using 200 mm (8 inch) wafer technology and fully automated production lines, it features narrow production tolerances and the highest product quality. This makes it suitable for a wide range of applications, from consumer electronics to industrial applications.

TOLL



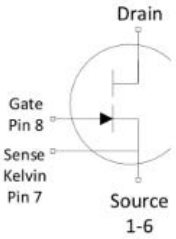
Features

- Enhancement mode transistor
- Ultra-fast switching
- No reverse-recovery charge
- Capable of reverse conduction
- Low gate and output charge
- Superior commutation ruggedness
- 2 kV HBM ESD standards

Benefits

- Normally OFF transistor technology ensures safe operation
- Enables rapid and precise power delivery control
- Improves system efficiency and reliability
- Ensures robust performance under challenging conditions

These features collectively make CoolGaN™ a game-changer in the realm of power conversion, offering a compelling combination of efficiency, compactness, and reliability.



Potential applications

Industrial, telecom, datacenter SMPS based on half-bridge hard and soft switching topologies such as totem pole PFC and high frequency LLC, as well as charger and adapter.

Product validation

Qualified according to relevant JEDEC tests.

Table 1 Key performance parameters

Parameter	Value	Unit
$V_{DS,max}$	650	V
$V_{DS,trans-max}$	900	V
$R_{DS(on),max}$	54	mΩ
$Q_{g,typ}$	6	nC
$I_{D,pulse}$	76	A
Q_{oss} @ 400 V	45	nC
Q_{rr}	0	nC

GaN Hal Pb RoHS

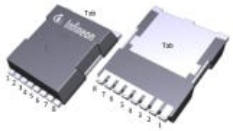
Public
IMT65R050M2H
Final datasheet

SiC MOSFET

CoolSiC™ MOSFET 650 V G2

Built on Infineon's robust 2nd generation Silicon Carbide trench technology, the 650 V CoolSiC™ MOSFET delivers unparalleled performance, superior reliability, and great ease of use. It enables cost effective, highly efficient, and simplified designs to fulfill the ever-growing system and market needs.

TOLL

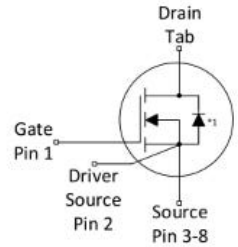


Features

- Ultra-low switching losses
- Benchmark gate threshold voltage, $V_{GS(th)}$ = 4.5 V
- Robust against parasitic turn-on even with 0 V turn-off gate voltage
- Flexible driving voltage and compatible with bipolar driving scheme
- Robust body diode operation under hard commutation events
- .XT interconnection technology for best-in-class thermal performance

Benefits

- Enables high efficiency and high power density designs
- Facilitates great ease of use and integration
- Provides the best price performance ratio compared to Industry's most ambitious roadmaps
- Reduces the size, weight and bill of materials of the systems
- Enhances system robustness and reliability



**1: Internal body diode*

Potential applications

- SMPS
- Solar PV inverters
- Energy storage and battery formation
- UPS
- EV charging infrastructure
- Motor drives

Product validation

Qualified according to relevant JEDEC tests.

Please note: The source and driver source pins are not exchangeable. Their exchange might lead to malfunction.

Table 1 Key performance parameters

Parameter	Value	Unit
V_{DSS} over full $T_{j,range}$	650	V
$R_{DS(on),typ}$	50	mΩ
$R_{DS(on),max}$	62	mΩ
$Q_{G,typ}$	22	nC
$I_{D,pulse}$	114	A
Q_{oss} @ 400 V	47	nC
E_{oss} @ 400 V	6.3	µJ

Hal Pb RoHS

IA30 original exhibit 5; source units and conditions discussed in text

Original article datasheet comparison. The 45 and 47 figures are Qoss in nanocoulombs. They are not capacitances. This exhibit is retained to make the unit correction auditable, not to endorse the author's subsequent ranking.

The May article's ROHM example has a similar topology problem: SCZ4006KTA is a **1200 V half-bridge**, not evidence that two 600 V switches sum their ratings to 1200 V. A half-bridge switch must block the relevant bus voltage while the other conducts. More generally, package selection, junction temperature, RMS current, gate drive, parasitic inductance, isolation, switching energy and protection have to be evaluated in the actual circuit. A hand-normalized table assembled from different datasheet test conditions is a screening aid, not a measured system comparison. [P5]

The opportunity survives those corrections. Wolfspeed's March 5 release introduced a commercially available 10 kV bare die for customer sampling and qualification. The current CPM3-10000-0300A page lists 305 mΩ typical resistance and 20 A continuous current under its stated conditions. Bare die still requires a customer's package, insulation, cooling, control and system qualification. The article's preliminary plots show resistance rising with junction temperature; its 3 kV capacitance and 6 kV switching test points are not a blanket recommended operating range. Wolfspeed separately describes 6 kV typical operation in its SST reliability paper. [P6, P7]

Absolute Maximum Ratings

Stress beyond those listed under absolute maximum ratings may damage the device.

Parameter	Symbol		Rating	Unit
Drain-Source Voltage, across T_{VJ}	$V_{DS(max)}$		10000	V
Maximum Gate-Source Voltage, Peak Transient Capability	$V_{GS(max)}$		-10/+25	V
Static Gate-Source Voltage	$V_{GS(op)}$		-5/+20	V
Continuous Drain Current[1]	I_{DS}	T_c = 25 °C	20	A
		T_c = 90 °C	15	
Pulsed Drain Current, t_p limited by $T_{VJ(max)}$	$I_{DS(pulse)}$		40	A
Virtual Junction and Storage Temperature	T_{VJ}, T_{stg}		-55 to +175	°C
Maximum Processing Temperature, in non-reactive ambient	T_{proc}		325	°C

Note 1. Assume testing at V_{GS} = +20 V, with a junction-to-case thermal resistance $R_{TH(J-C)}$ = 0.34 °C/W.

IA26 original exhibit 45; source units and conditions discussed in text

Saved preliminary 10 kV ratings: 20 A at a 25°C case temperature falls to 15 A at 90°C under the stated thermal assumptions. These are device limits, not converter output or shipment guidance. Additional curves remain attributed originals; the independently retrieved public datasheet is abbreviated.

Wolfspeed also introduced 3.3 kV module families on May 21. Its Amperesand case study describes that voltage class in a modular SST architecture. This is useful counter-evidence to a screen that excludes modules and assumes the highest-voltage die automatically wins. Fewer conversion stages can help, while redundancy, serviceability, insulation, yield and cost can favor a different topology. No identified exclusivity, quantity or price in these sources establishes a new Wolfspeed revenue layer. [P8, P9]

Probability, recognition and overlap. Assign a **60% analyst probability** that the 10 kV/SST opportunity contributes identifiable Wolfspeed device sales by calendar 2028. A deliberately bounded attribution diagnostic is **1-5% of the existing $370M CY2028 AI-device forecast**, or **$3.7-18.5M conditional revenue**. This range is a low-confidence model-composition assumption, not a reported order or a new full-system TAM. Probability weighting gives $2.22-11.10M of potential contribution inside that existing forecast; the overlap adjustment is **100%**, so no extra revenue is added. The existing high-voltage AI vector already grows from $134M in 2027 to $370M in 2028 and includes conversion/protection content. A second SST sales line would count some of the same demand twice.

At the current 2028 AI gross margins of approximately 28.8-33.2%, the conditional slice corresponds to roughly **$2.47-13.17M of manufacturing cost and $1.07-6.15M of GP** before attributable engineering and selling costs. These are diagnostic bounds, not a second forecast; system packaging, test and qualification spending can consume the gross contribution. The existing consolidated opex schedule remains in place. Incremental revenue, COGS, opex and OP are each **zero** after overlap. No price premium or margin upgrade is inferred from the erroneous transistor comparison, and no 10 kV supplier invoice is confused with the value of an entire SST.

The next useful evidence is a named qualified customer, accepted die/module shipments, actual Wolfspeed content per delivered system, sustainable yield and a realized price/margin bridge. Qualification or a design win can support a nonzero probability before official volume guidance; it cannot establish the missing unit economics. Failure or delay would be assessed against the already ambitious AI ramp, rather than simply removing an unmodeled extra market.

The engineering evidence leaves operating revenue and costs unchanged. The later capital-share update in this full artifact changes only dilution. The negative active-window result remains a lack of positive earnings support, not a forecast of a negative quoted price. Analytical Base 58 and current Excel Base 58 remain unchanged.

Engineering sources: P1: Irrational Analysis, April 25, 2026; P2: Irrational Analysis, May 17, 2026; P3: Infineon IGT65R045D2 datasheet; P4: Infineon MOSFET datasheet explanation, section 2.3; P5: ROHM 1200 V half-bridge product; P6: Wolfspeed 10 kV introduction, March 5; P7: Wolfspeed SST reliability discussion, March 6; P8: Wolfspeed 3.3 kV modules, May 21; P9: Amperesand case study.

1. Supercycle, supply chain and durable advantage

Higher-voltage power architectures increase the scope for silicon carbide in power supplies, protection and transformers. Wolfspeed has an operating200mm device platform and a qualifiedLITEON relationship. The investment case still requires greater factory loading and profitable volume, with substantial dilution and legacy debt.

Power-system demand → qualifiedSiC devices/materials → shipped units×content → factory absorption andGP →R&D/admin/restructuring → operating earnings → dilution and debt obligations. Hyperscaler construction budgets cannot be treated as semiconductor sales.

Business / curve	Stage	Evidence and next inflection
AI power devices	Early commercial growth	Small base; qualification precedes significant shipments.
EV/industrial devices	Recovery and technology transition	Gen5 may improve competitiveness, demand is uncertain.

Business / curve	Stage	Evidence and next inflection
External materials	Transition	150mmLTAs wind down while200mm qualification continues.

Stage percentages are qualitative framework bands relative to a possible mature opportunity, not measured market penetration. A current capacity constraint can coexist with a larger long-term TAM; qualification and economic capture decide how much reaches this company.

2. Forward TAM and the three-phase expansion lens

Phase	Engineering opportunity	Demand-to-company bridge
2026-27	Utilization before profitability	Q32026 revenue150M matches fiscalQ1 midpoint;GP approaches zero near800M annual run rate.
2028-29	AI power conversion scales	AI infrastructure gains share of device revenue while existing markets recover.
2030+	Content-based capture	AI annual1240M could represent100000high-power rack/system equivalents×12400USD of WolfspeedSiC content across power/protection stages. Other devices988M could represent2Mvehicle/industrial equivalents×494USD. External materials428M could represent428000wafer-equivalents×1000USD. Counts,content andprices are analyst assumptions; there is no disclosed order bridge to these amounts.

The 100% / 50% / 20% thematic weights guide thesis assessment. They are not blanket discounts to reported sales, contracts or all forecasts. Programme probabilities below govern only separately identified optionality. Current reported sales calibrate the starting point; they do not cap the future market.

3. Current earnings and management evidence

FiscalQ4 endedJune28 revenue149.6M is power106.3M plus materials43.3M. Cost186.4M producesGP-36.8M. R&D27.9M plusSG&A37.3M gives coreOP-102.0M;15.9M restructuring less2.5M disposal gain takes reportedOP to-115.4M [E1]. Negative consolidatedGM reflects fixed-cost underutilization and does not prove that each incremental unit has negative contribution.

Full May5 and August19 management earnings remarks andQ&As were reviewed [E3/E4]. AI revenue grew approximately30% sequentially inQ3 and20% inQ4 from a moderate base. The CFO estimates gross-margin breakeven near800M annual revenue, mix dependent; this is notEBITDA orGAAPOP breakeven. Auto and150mm material transition remain uncertain.

Both latest full management transcripts and theannual filing were reviewed. Transcript wording for wafer sizes and currencies contains errors; source financial statements take precedence. There is no quantifiedAI segment or published product-levelGAAP margin.

4. Local evidence and calibration packet

Historical reported product revenue andGAAPGP reconcile exactly. Internal substrates used in Wolfspeed devices are excluded from external-material revenue to prevent double counting. No verified capex-to-Wolfspeed capture coefficient is available; the long-term system content bridge isLOW confidence.

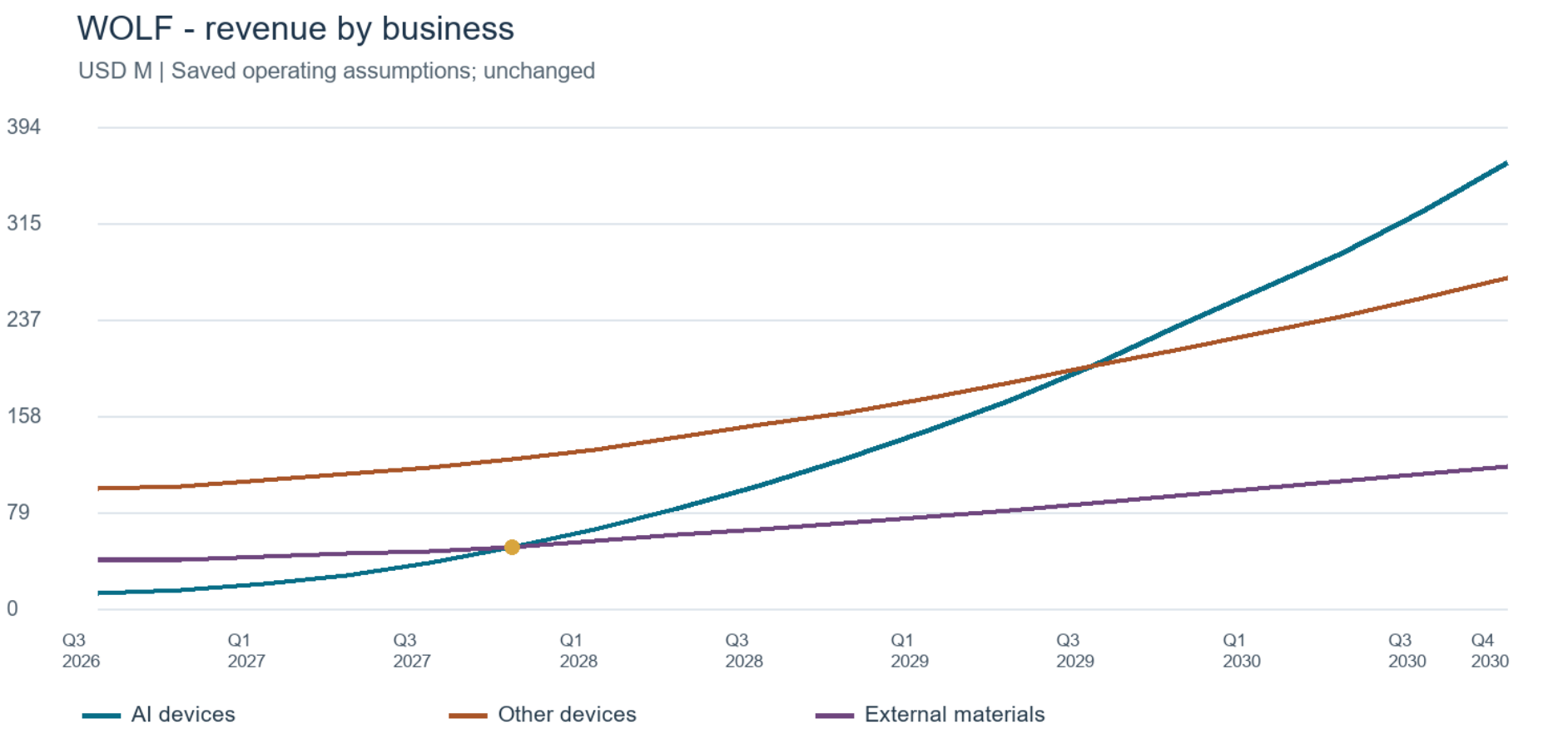
5. Summing thesis score

Category	Points / maximum	Evidence
Core thesis	13/20	High-voltage technology and vertical integration, contested markets.
Inflection / cycle	5/10	Device growth offsets material weakness; margins remain negative.
Forward TAM	8/10	AI power and transformer applications enlarge addressable uses.
Catalysts	4/5	LITEON qualification andGen5 deployment.
Macro / policy	7/10	Electrification and domestic supply needs.
Alpha / evidence	3/5	Profitability gate and full dilution clarified.
Risk deductions	-5	Five distinct risk classes.

Sum: 35/60 × 100 = 58.333333%, conventionally rounded to 58%. This advisory thesis score does not replace any portfolio Base override. The separate risk classes below total the displayed deduction; holding status and classification do not waive that deduction.

Financial views of the current model

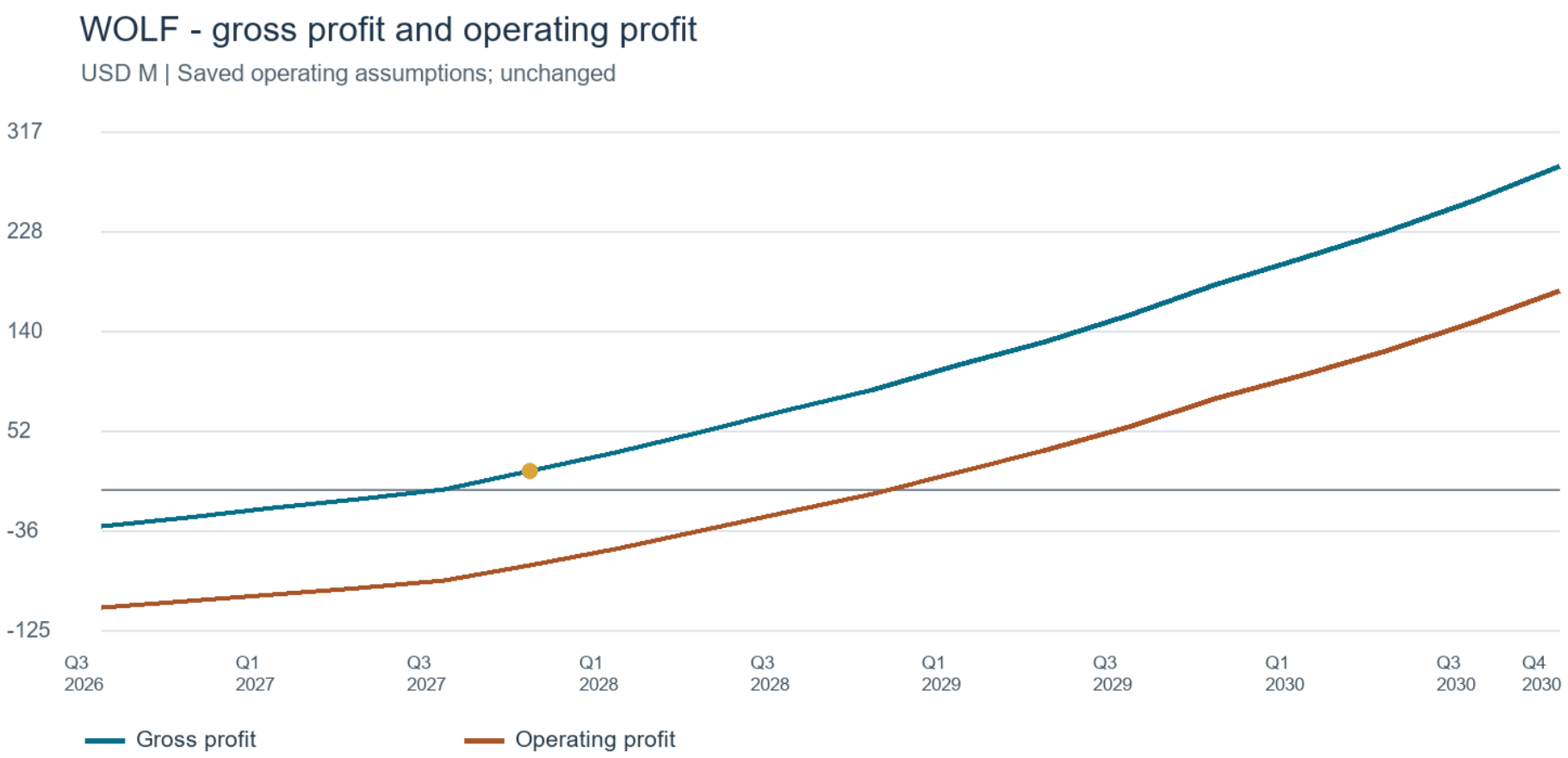
These views use the existing quarterly assumptions and valuation convention. Financial values are USD millions except per-share ceilings and percentages. Gold identifies the active Q4 2027 point on the first series where shown; it is not a forecast change. Underlying figures are reproduced below each chart.



WOLF - revenue by business

Quarter	AI devices	Other devices	External materials
Q3 2026	12.000	98.000	40.000
Q4 2026	15.000	100.000	40.000
Q1 2027	20.000	105.000	42.000
Q2 2027	27.000	110.000	45.000
Q3 2027	37.000	115.000	47.000

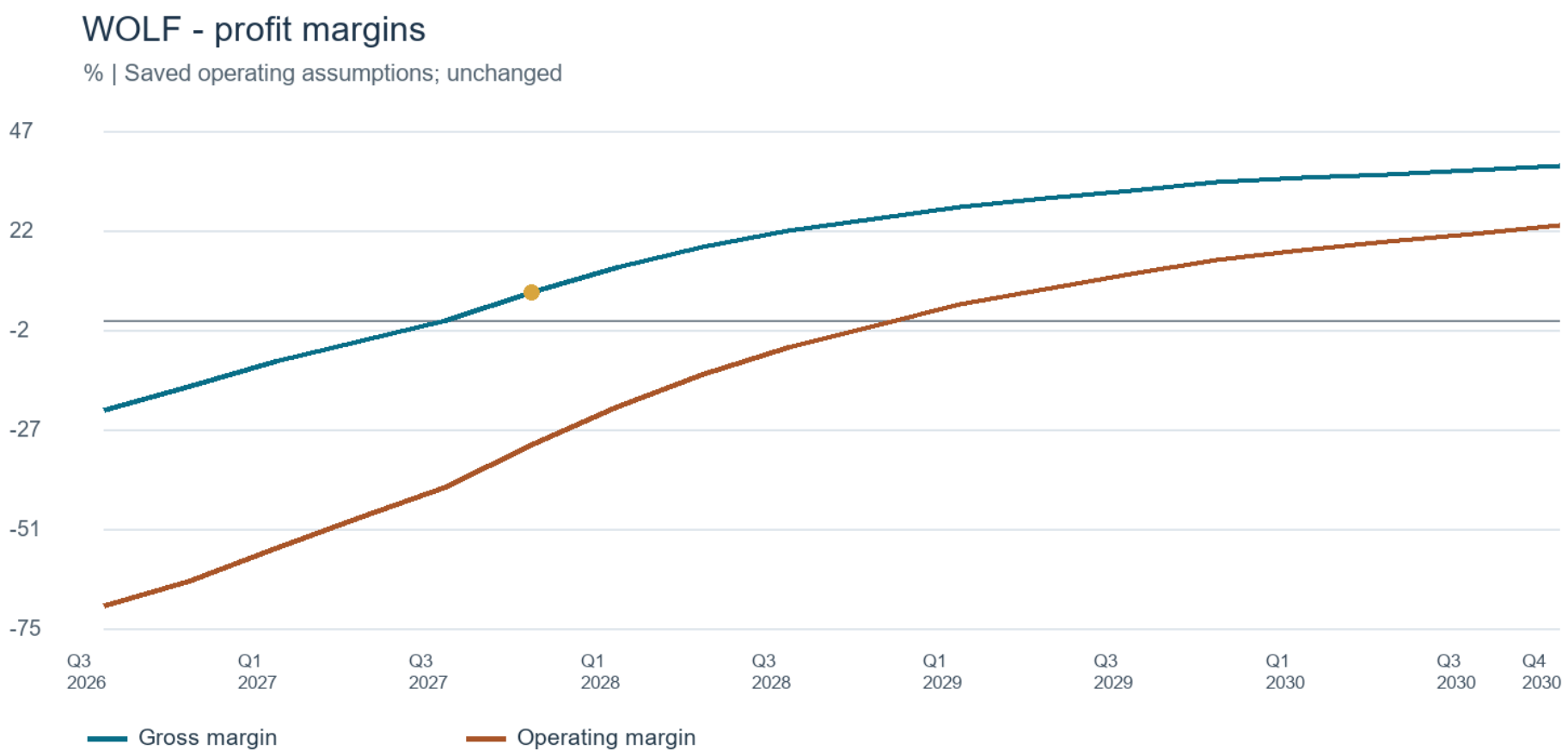
Quarter	AI devices	Other devices	External materials
Q4 2027	50.000	122.000	50.000
Q1 2028	65.000	130.000	55.000
Q2 2028	82.000	140.000	60.000
Q3 2028	101.000	150.000	65.000
Q4 2028	122.000	160.000	70.000
Q1 2029	145.000	172.000	75.000
Q2 2029	170.000	185.000	80.000
Q3 2029	198.000	198.000	86.000
Q4 2029	230.000	211.000	92.000
Q1 2030	260.000	225.000	98.000
Q2 2030	290.000	239.000	104.000
Q3 2030	325.000	254.000	110.000
Q4 2030	365.000	270.000	116.000



WOLF - gross profit and operating profit

Quarter	Gross profit	Operating profit
Q3 2026	-33.000	-105.000
Q4 2026	-24.800	-99.035
Q1 2027	-16.700	-93.171
Q2 2027	-9.100	-87.806
Q3 2027	0.000	-80.941

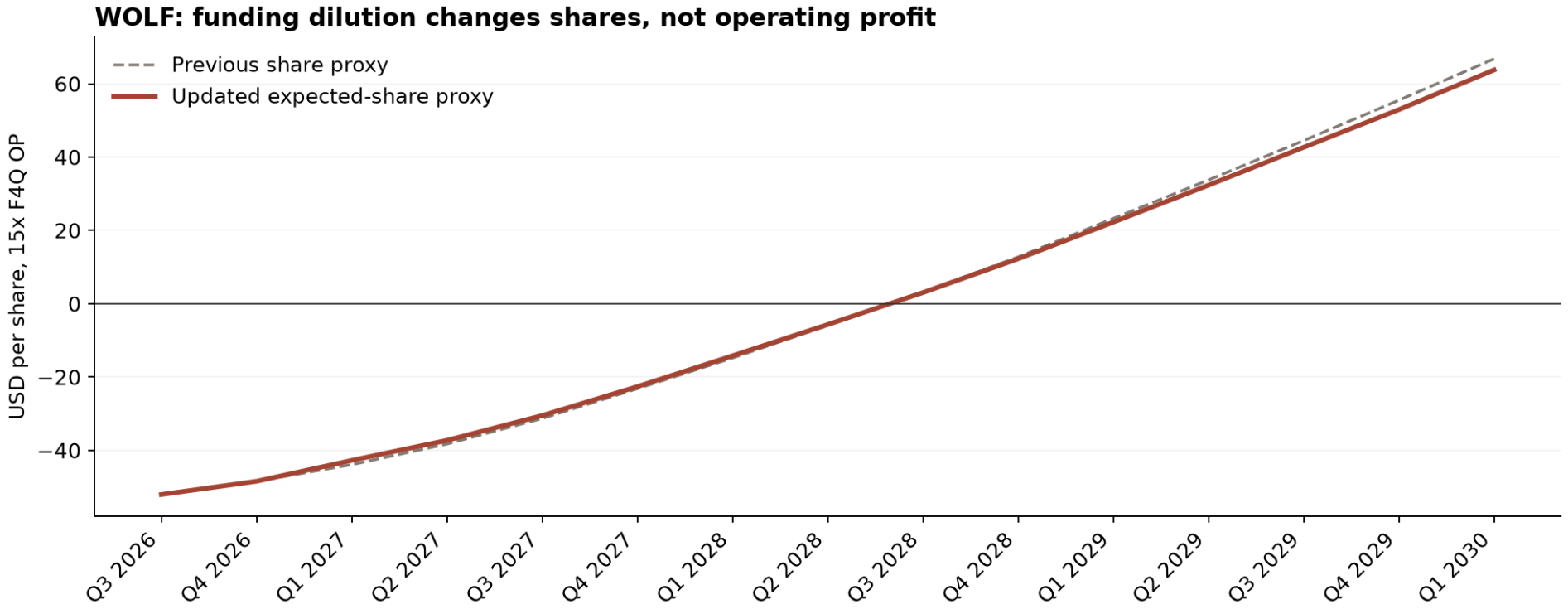
Quarter	Gross profit	Operating profit
Q4 2027	15.540	-67.636
Q1 2028	32.500	-52.912
Q2 2028	50.760	-36.887
Q3 2028	69.520	-20.362
Q4 2028	88.000	-4.118
Q1 2029	109.760	15.407
Q2 2029	130.500	33.912
Q3 2029	154.240	55.416
Q4 2029	181.220	80.161
Q1 2030	204.050	100.756
Q2 2030	227.880	122.351
Q3 2030	254.930	147.165
Q4 2030	285.380	175.380



WOLF - profit margins

Quarter	Gross margin	Operating margin
Q3 2026	-22.000	-70.000
Q4 2026	-16.000	-63.894
Q1 2027	-10.000	-55.791
Q2 2027	-5.000	-48.245
Q3 2027	0.000	-40.674

Quarter	Gross margin	Operating margin
Q4 2027	7.000	-30.467
Q1 2028	13.000	-21.165
Q2 2028	18.000	-13.081
Q3 2028	22.000	-6.444
Q4 2028	25.000	-1.170
Q1 2029	28.000	3.930
Q2 2029	30.000	7.796
Q3 2029	32.000	11.497
Q4 2029	34.000	15.040
Q1 2030	35.000	17.282
Q2 2030	36.000	19.329
Q3 2030	37.000	21.359
Q4 2030	38.000	23.353

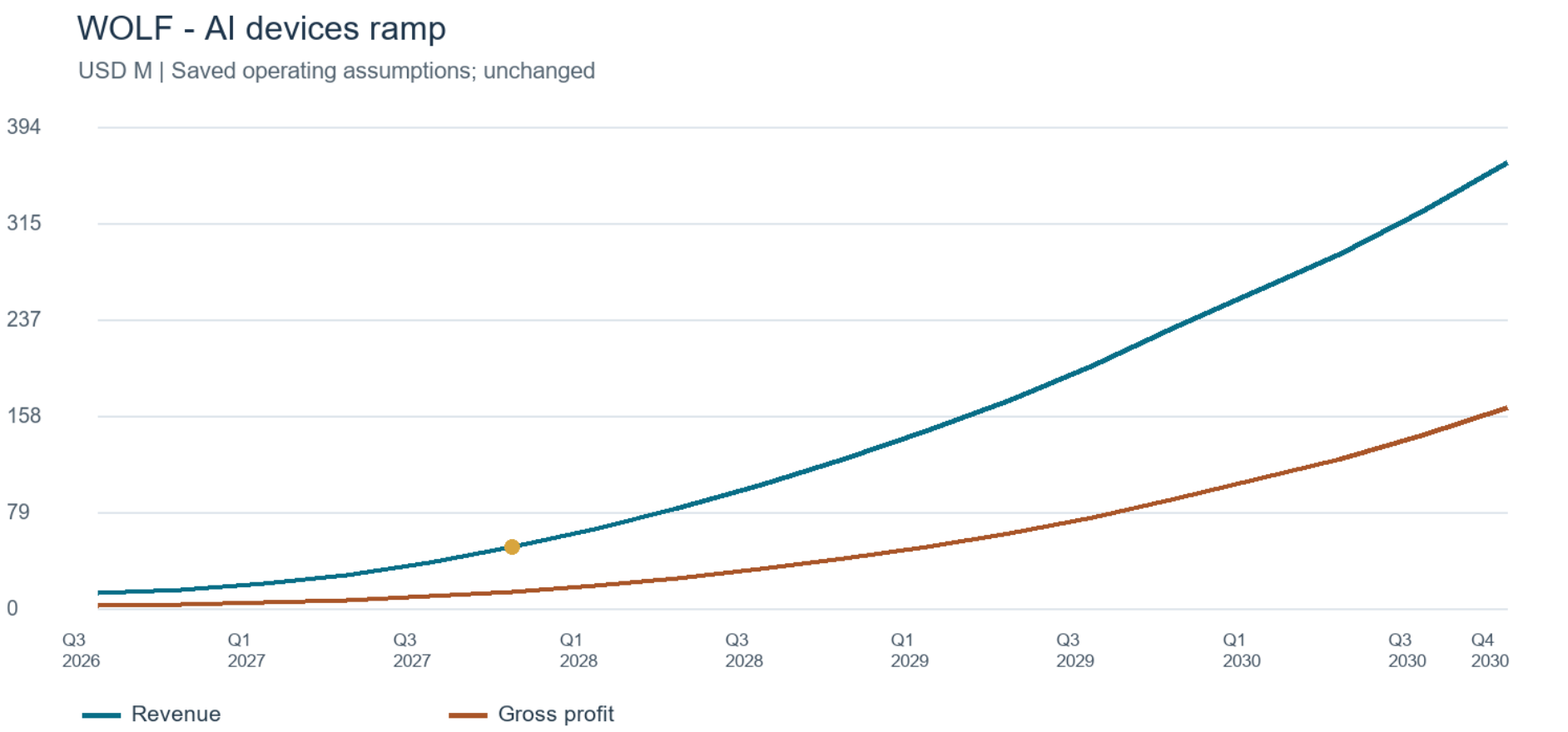


WOLF - quarterly model ceiling; negative values are formula diagnostics, not negative market-price forecasts

Quarter	Forward OP valuation
Q3 2026	-52.028617
Q4 2026	-48.418201
Q1 2027	-42.716247
Q2 2027	-37.254463
Q3 2027	-30.499255
Q4 2027	-22.602623
Q1 2028	-14.190068
Q2 2028	-5.671369
Q3 2028	3.046105

Quarter	Forward OP valuation
Q4 2028	12.263390
Q1 2029	22.264044
Q2 2029	32.344801
Q3 2029	42.672219
Q4 2029	52.957471
Q1 2030	63.774267

Negative OP-derived ceilings indicate no positive earnings support from this method in that quarter; they are not forecasts of negative traded share prices.



WOLF - AI devices ramp

Quarter	Revenue	Gross profit
Q3 2026	12.000	2.400
Q4 2026	15.000	3.221
Q1 2027	20.000	4.588
Q2 2027	27.000	6.591
Q3 2027	37.000	9.576

Quarter	Revenue	Gross profit
Q4 2027	50.000	13.676
Q1 2028	65.000	18.735
Q2 2028	82.000	24.841
Q3 2028	101.000	32.082
Q4 2028	122.000	40.547
Q1 2029	145.000	50.324
Q2 2029	170.000	61.500
Q3 2029	198.000	74.541
Q4 2029	230.000	89.971
Q1 2030	260.000	105.529
Q2 2030	290.000	121.971
Q3 2030	325.000	141.471
Q4 2030	365.000	164.250

6. Revenue sources, capture and product margins

Split the reported106.3M power line into10M illustrativeAI and96.3M other power; separateAI sales are undisclosed. Assign historicalAI10% and materials-40% GM; solve other powerGM=(-36.8-1+17.32)/96.3=-21.2669% to reproduceGAAPGP. Forward product margins jointly reflect utilization: consolidatedGM improves from-22% to zero near199M quarterly revenue, then38% by2030. The solved other-device margin path is an analyst utilization assumption, not an issuer segment forecast.

Vector	Actual calibration / source	Forward demand and capture	Margin derivation / constraint
AI power	10M allocated within106.3M power	2030 annual1240M	20→45% futureGM; requires volume and competitive content.
Other devices	96.3M residual	2030 annual988M	Margin path solves explicit utilization bridge.
External materials	43.3M reported	2030 annual428M	-35→25% GM as demand absorbs fixed costs.

Confidence: reported historical totals are HIGH. Source-based forward extrapolations are MEDIUM only where the cited source sizes the relevant period; other product allocations, future capture and margin assumptions are LOW. A source for the historical anchor does not make an unsourced future factor management guidance. The detailed factors below are the single modeled base case, with uncertain assumptions exposed for review.

V1. AI infrastructure power devices

Sources: E1/E3/E5; explicit allocatedAI anchor. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2026	10.000000	1.200000	12.000	20.000%	2.400
Q4 2026	10.000000	1.500000	15.000	21.471%	3.221
Q1 2027	10.000000	2.000000	20.000	22.941%	4.588
Q2 2027	10.000000	2.700000	27.000	24.412%	6.591
Q3 2027	10.000000	3.700000	37.000	25.882%	9.576
Q4 2027	10.000000	5.000000	50.000	27.353%	13.676
Q1 2028	10.000000	6.500000	65.000	28.824%	18.735
Q2 2028	10.000000	8.200000	82.000	30.294%	24.841
Q3 2028	10.000000	10.100000	101.000	31.765%	32.082
Q4 2028	10.000000	12.200000	122.000	33.235%	40.547
Q1 2029	10.000000	14.500000	145.000	34.706%	50.324
Q2 2029	10.000000	17.000000	170.000	36.176%	61.500
Q3 2029	10.000000	19.800000	198.000	37.647%	74.541
Q4 2029	10.000000	23.000000	230.000	39.118%	89.971
Q1 2030	10.000000	26.000000	260.000	40.588%	105.529
Q2 2030	10.000000	29.000000	290.000	42.059%	121.971
Q3 2030	10.000000	32.500000	325.000	43.529%	141.471
Q4 2030	10.000000	36.500000	365.000	45.000%	164.250

V2. Other power devices

Sources: E1; power total lessAI allocation. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2026	96.300000	1.017653	98.000	-21.837%	-21.400
Q4 2026	96.300000	1.038422	100.000	-15.432%	-15.432
Q1 2027	96.300000	1.090343	105.000	-9.098%	-9.553
Q2 2027	96.300000	1.142264	110.000	-4.278%	-4.706
Q3 2027	96.300000	1.194185	115.000	0.207%	0.238
Q4 2027	96.300000	1.266874	122.000	8.639%	10.540
Q1 2028	96.300000	1.349948	130.000	16.437%	21.368
Q2 2028	96.300000	1.453790	140.000	22.925%	32.095
Q3 2028	96.300000	1.557632	150.000	27.890%	41.835
Q4 2028	96.300000	1.661475	160.000	31.074%	49.718
Q1 2029	96.300000	1.786085	172.000	34.428%	59.216
Q2 2029	96.300000	1.921080	185.000	35.644%	65.941
Q3 2029	96.300000	2.056075	198.000	37.058%	73.375
Q4 2029	96.300000	2.191070	211.000	38.501%	81.238
Q1 2030	96.300000	2.336449	225.000	37.510%	84.397
Q2 2030	96.300000	2.481828	239.000	36.507%	87.251
Q3 2030	96.300000	2.637591	254.000	35.371%	89.842
Q4 2030	96.300000	2.803738	270.000	34.122%	92.130

V3. External materials

Sources: E1; reported external material sales. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2026	43.300000	0.923788	40.000	-35.000%	-14.000
Q4 2026	43.300000	0.923788	40.000	-31.471%	-12.588
Q1 2027	43.300000	0.969977	42.000	-27.941%	-11.735
Q2 2027	43.300000	1.039261	45.000	-24.412%	-10.985
Q3 2027	43.300000	1.085450	47.000	-20.882%	-9.815
Q4 2027	43.300000	1.154734	50.000	-17.353%	-8.676
Q1 2028	43.300000	1.270208	55.000	-13.824%	-7.603
Q2 2028	43.300000	1.385681	60.000	-10.294%	-6.176
Q3 2028	43.300000	1.501155	65.000	-6.765%	-4.397
Q4 2028	43.300000	1.616628	70.000	-3.235%	-2.265
Q1 2029	43.300000	1.732102	75.000	0.294%	0.221
Q2 2029	43.300000	1.847575	80.000	3.824%	3.059
Q3 2029	43.300000	1.986143	86.000	7.353%	6.324
Q4 2029	43.300000	2.124711	92.000	10.882%	10.012
Q1 2030	43.300000	2.263279	98.000	14.412%	14.124
Q2 2030	43.300000	2.401848	104.000	17.941%	18.659
Q3 2030	43.300000	2.540416	110.000	21.471%	23.618
Q4 2030	43.300000	2.678984	116.000	25.000%	29.000

7. Operating-cost bridge and quarterly operating model

FiscalQ1 nonGAAPOpex guide62-66M; model64M midpoint plus4M recurring compensation/GAAP differences plus4M restructuring allowance=72M. Total quarterly costs rise to110M. Product gross margins retain depreciation and factory costs. No old claim of800MEBITDA breakeven is used: the latestCFO discussion refers toGP breakeven.

Profit basis: GAAP-style operating earnings including depreciation and stock compensation; no investment gain or financing income. Amounts below are USD millions.

Quarter	Status	Revenue	GP / contribution	Opex %	Opex	OP margin	Operating profit	Main driver	Confidence
Q3 2026	Model	150.000	-33.000	48.000%	72.000	-70.000%	-105.000 (neg)	FiscalQ1 guidance midpoint	LOW model
Q4 2026	Model	155.000	-24.800	47.894%	74.235	-63.894%	-99.035 (neg)	Factory loading	LOW model
Q1 2027	Model	167.000	-16.700	45.791%	76.471	-55.791%	-93.171 (neg)	AI power qualifications	LOW model
Q2 2027	Model	182.000	-9.100	43.245%	78.706	-48.245%	-87.806 (neg)	Device demand recovery	LOW model
Q3 2027	Model	199.000	0.000	40.674%	80.941	-40.674%	-80.941 (neg)	ApproximateGP breakeven	LOW model
Q4 2027	Model	222.000	15.540	37.467%	83.176	-30.467%	-67.636 (neg)	Broader high-voltage demand	LOW model
Q1 2028	Model	250.000	32.500	34.165%	85.412	-21.165%	-52.912 (neg)	External200mm transition	LOW model
Q2 2028	Model	282.000	50.760	31.081%	87.647	-13.081%	-36.887 (neg)	Higher factory utilization	LOW model
Q3 2028	Model	316.000	69.520	28.444%	89.882	-6.444%	-20.362 (neg)	AI system shipments	LOW model
Q4 2028	Model	352.000	88.000	26.170%	92.118	-1.170%	-4.118 (neg)	Renewal of material demand	LOW model
Q1 2029	Model	392.000	109.760	24.070%	94.353	3.930%	15.407	Device volume	LOW model
Q2 2029	Model	435.000	130.500	22.204%	96.588	7.796%	33.912	Product mix	LOW model
Q3 2029	Model	482.000	154.240	20.503%	98.824	11.497%	55.416	Higher-voltage penetration	LOW model
Q4 2029	Model	533.000	181.220	18.960%	101.059	15.040%	80.161	Manufacturing yield	LOW model
Q1 2030	Model	583.000	204.050	17.718%	103.294	17.282%	100.756	MoreAI power content	LOW model
Q2 2030	Model	633.000	227.880	16.671%	105.529	19.329%	122.351	Scaled device demand	LOW model

Quarter	Status	Revenue	GP / contribution	Opex %	Opex	OP margin	Operating profit	Main driver	Confidence
Q3 2030	Model	689.000	254.930	15.641%	107.765	21.359%	147.165	Cost discipline	LOW model
Q4 2030	Model	751.000	285.380	14.647%	110.000	23.353%	175.380	Mature utilization scenario	LOW model

OP = sum of product GP/contribution - explicit operating costs. OP margin = OP ÷ revenue. Displayed numbers are rounded for reading; the workbook calculates from unrounded inputs. Guidance assigned to a future quarter retains Model status because guidance is not an actual result.

8. Multiple, shares and recognition convention

Retain15× operating earnings. This method is not a debt-adjusted equityDCF and does not capitalize cash receipts or subtract net debt; capital-structure risk remains explicit. Negative early price targets identify a lack of earnings-method support.

The retained June gross converted bridge is 110.860M shares: 52.162M issued plus 21.836M, 11.096M and 18.822M reserved conversion shares, 4.944M Renesas warrants and 2M prefunded warrants. The prior model rounded to 111M and rose to 125M to cover awards and capital. The 3.766M outstanding awards and 8.332M ungranted future capacity sum to 12.098M of that 14M increase, leaving 1.902M as the generic capital reserve. October 11 financing dilution offsets that reserve quarter by quarter before adding new shares. Existing conversion shares are never added again. The September 11 proxy gives 52.930569M actual common shares at August 31, a different denominator from potential converted shares; the earlier August 13 snapshot was 52.995396M. The 1.7M ESPP proposal awaiting the October 27 vote is not issued stock. No warrant exercise cash is credited.

The valuation denominator is the price target quarter’s diluted-share proxy. It is not a historical EPS weighted-average assumption disguised as a fully known future capitalization. Unit factor: 1; this converts the model units to USD per share. No cash balance, enterprise-value adjustment or after-tax earnings substitution is added to the operating-profit price target.

9. Complete rolling four-quarter price target derivation

price target quarter	Exact earnings window	Forward 4Q OP	Multiple	Diluted shares M	price target USD	Key inflection
Q3 2026	Q3 2026 → Q2 2027	-385.011765	15×	111.000000	-52.028617 (neg)	FiscalQ1 guidance midpoint
Q4 2026	Q4 2026 → Q3 2027	-360.952941	15×	111.823529	-48.418201 (neg)	Factory loading
Q1 2027	Q1 2027 → Q4 2027	-329.554118	15×	115.724393	-42.716247 (neg)	AI power qualifications
Q2 2027	Q2 2027 → Q1 2028	-289.295294	15×	116.480793	-37.254463 (neg)	Device demand recovery
Q3 2027	Q3 2027 → Q2 2028	-238.376471	15×	117.237193	-30.499255 (neg)	ApproximateGP breakeven
Q4 2027	Q4 2027 → Q3 2028	-177.797647	15×	117.993593	-22.602623 (neg)	Broader high-voltage demand
Q1 2028	Q1 2028 → Q4 2028	-114.278824	15×	120.801558	-14.190068 (neg)	External200mm transition
Q2 2028	Q2 2028 → Q1 2029	-45.960000	15×	121.557958	-5.671369 (neg)	Higher factory utilization
Q3 2028	Q3 2028 → Q2 2029	24.838824	15×	122.314358	3.046105	AI system shipments
Q4 2028	Q4 2028 → Q3 2029	100.617647	15×	123.070758	12.263390	Renewal of material demand
Q1 2029	Q1 2029 → Q4 2029	184.896471	15×	124.570678	22.264044	Device volume
Q2 2029	Q2 2029 → Q1 2030	270.245294	15×	125.327078	32.344801	Product mix
Q3 2029	Q3 2029 → Q2 2030	358.684118	15×	126.083478	42.672219	Higher-voltage penetration
Q4 2029	Q4 2029 → Q3 2030	450.432941	15×	127.583398	52.957471	Manufacturing yield
Q1 2030	Q1 2030 → Q4 2030	545.651765	15×	128.339798	63.774267	MoreAI power content

- **Q3 2026:** (-105.000000 + -99.035294 + -93.170588 + -87.805882) × 15 × 1 ÷ 111.000000 = **-52.028617 USD/share**. Window: Q3 2026 through Q2 2027.
- **Q4 2026:** (-99.035294 + -93.170588 + -87.805882 + -80.941176) × 15 × 1 ÷ 111.823529 = **-48.418201 USD/share**. Window: Q4 2026 through Q3 2027.
- **Q1 2027:** (-93.170588 + -87.805882 + -80.941176 + -67.636471) × 15 × 1 ÷ 115.724393 = **-42.716247 USD/share**. Window: Q1 2027 through Q4 2027.
- **Q2 2027:** (-87.805882 + -80.941176 + -67.636471 + -52.911765) × 15 × 1 ÷ 116.480793 = **-37.254463 USD/share**. Window: Q2 2027 through Q1 2028.
- **Q3 2027:** (-80.941176 + -67.636471 + -52.911765 + -36.887059) × 15 × 1 ÷ 117.237193 = **-30.499255 USD/share**. Window: Q3 2027 through Q2 2028.
- **Q4 2027:** (-67.636471 + -52.911765 + -36.887059 + -20.362353) × 15 × 1 ÷ 117.993593 = **-22.602623 USD/share**. Window: Q4 2027 through Q3 2028.
- **Q1 2028:** (-52.911765 + -36.887059 + -20.362353 + -4.117647) × 15 × 1 ÷ 120.801558 = **-14.190068 USD/share**. Window: Q1 2028 through Q4 2028.
- **Q2 2028:** (-36.887059 + -20.362353 + -4.117647 + 15.407059) × 15 × 1 ÷ 121.557958 = **-5.671369 USD/share**. Window: Q2 2028 through Q1 2029.
- **Q3 2028:** (-20.362353 + -4.117647 + 15.407059 + 33.911765) × 15 × 1 ÷ 122.314358 = **3.046105 USD/share**. Window: Q3 2028 through Q2 2029.

- **Q4 2028:** (-4.117647 + 15.407059 + 33.911765 + 55.416471) × 15 × 1 ÷ 123.070758 = **12.263390 USD/share**. Window: Q4 2028 through Q3 2029.
- **Q1 2029:** (15.407059 + 33.911765 + 55.416471 + 80.161176) × 15 × 1 ÷ 124.570678 = **22.264044 USD/share**. Window: Q1 2029 through Q4 2029.
- **Q2 2029:** (33.911765 + 55.416471 + 80.161176 + 100.755882) × 15 × 1 ÷ 125.327078 = **32.344801 USD/share**. Window: Q2 2029 through Q1 2030.
- **Q3 2029:** (55.416471 + 80.161176 + 100.755882 + 122.350588) × 15 × 1 ÷ 126.083478 = **42.672219 USD/share**. Window: Q3 2029 through Q2 2030.
- **Q4 2029:** (80.161176 + 100.755882 + 122.350588 + 147.165294) × 15 × 1 ÷ 127.583398 = **52.957471 USD/share**. Window: Q4 2029 through Q3 2030.
- **Q1 2030:** (100.755882 + 122.350588 + 147.165294 + 175.380000) × 15 × 1 ÷ 128.339798 = **63.774267 USD/share**. Window: Q1 2030 through Q4 2030.

10. Annual trajectory cross-check

Calendar year	Revenue	GP / contribution	Opex	OP	OP margin
2027	770.000	-10.260	319.294	-329.554	-42.799%
2028	1,200.000	240.780	355.059	-114.279	-9.523%
2029	1,842.000	575.720	390.824	184.896	10.038%
2030	2,656.000	972.240	426.588	545.652	20.544%

These are calendar-year sums of the quarterly path, not extra revenue and not the live valuation method. For non-December fiscal reporters, fiscal guidance and these calendar totals must be aligned using the period mapping above; a four-quarter run rate is not automatically the same fiscal year. Outer-year outputs are conditional TAM/capture trajectories, not verified management sales targets.

CalendarQ32026 aligns approximately with fiscalQ1FY2027Jul-Sep; the52/53-week fiscal calendar creates a few-day boundary difference. CY2030 sum1240+988+428=2656M. This requires substantial execution beyond current600M annualized sales and remains a conditional capture case.

11. Contribution price sensitivity, unchanged operating forecasts

Only the assumed contribution equity issue price changes. Probabilities, warrant basis, draw dates, operating profit and the old reserve offset stay fixed. No warrant exercise price is inferred from this table.

Equity price proxy USD	Q4 2027 shares M	Q4 2027 formula result USD	Q1 2030 positive ceiling USD
24.464	118.238852	-22.555739	63.491158
30.580	117.993593	-22.602623	63.774267
36.696	117.830088	-22.633987	63.964412

A lower issuance price means more dilution. The negative near-window result becomes less negative mathematically, while the positive later ceiling falls. Neither effect is an operating forecast revision.

12. Programme and probability ledger

Programme / signal	Source and probability	Amount / unit base	Treatment in this model	Escalation / recognition
LITEON qualification	E5;HIGH technical milestone	No minimum purchase disclosed	Included once inAI device ramp	Monitor commercial shipments
OtherAI design wins	E3;HIGH qualitative	MacMic and ecosystem engagement	No invented hyperscaler commitment	Require revenue disclosure
GE AerospaceMoU	E1/E3;HIGH agreement	No guaranteed purchase volume	Optional applications within device forecast	Track qualification andorders
300mm beyond power	E4;early opportunity	No verified scaled revenue	No separate add-on inbase case	Require named commercial program

Probabilities are analyst judgments, not measured frequencies. Embedded revenue is not multiplied or added a second time. Under Framework 8.0, new credible evidence receives an explicit bounded incremental bridge before confirmation; a zero contribution needs an overlap, timing, counter-evidence or genuinely unquantifiable-after-proxy explanation. The October 6 section states that disposition for the evidence reviewed here. Existing unrelated program forecasts are preserved.

13. Scored risk classes

Risk class	Scored deduction	Mechanism and consequence
Factory execution	-1	Utilization and yields determine margin recovery.
Demand transition	-1	EV and150-to200mm timing may disappoint.
Competition and price	-1	AlternativeSiC andGaN/silicon solutions constrain capture.
Debt and liquidity	-1	High financing costs and cash burn remain.
Dilution	-1	Convertible notes, warrants and equity awards materially expand shares.

14. Dated catalysts, counters and escalation

1. **Q32026-2027 - Gross-margin gate** Revenue, mix and utilization versus roughly800Mrun-rate reference.
2. **2027-28 - Qualified demand** AI device sales and200mm material conversion.
3. **2028-30 - Cash and capital** Debt refinancing, share issuance and sustainedGAAP profitability.

15. Cross-book monitoring

LITEON power systems can generateSiC demand; GaN andsilicon can serve other stages. Soitec andotherSiC suppliers face the same demand/pricing cycle with different products. A broad800VDC rollout benefits multiple suppliers and is not an exclusiveWolfspeed allocation.

16. Source register and model limits

E1 - Wolfspeed fiscalQ4 results (2026-08-19). Financial statements and fiscalQ1 revenue/expense guidance.

E2 - Fiscal2026 Form10-K (2026-08-20). Share reserve table, debt, fresh-start accounting and product lines.

E3 - FiscalQ4 full management transcript (2026-08-19). Gross-margin breakeven estimate and end-market questions.

E4 - FiscalQ3 full management transcript (2026-05-05). Go-to-market and high-voltage opportunity; financial transcription errors checked against filings.

E5 - LITEON800VDC collaboration (2026-08-06). Successful qualification in sidecar and compute-rackPSU platforms, no purchase minimum.

No numericalAI revenue amount or hyperscaler market share is disclosed. The10M historicalAI allocation is only a calibration device. The model does not assume thatSiC replaces all low-voltageGaN or silicon stages, or that300mm packaging has reached production-scale revenue.

Workbook: v3_2_master_portfolio.xlsx, model tab WOLF_Q_Model. It owns portfolio ranking, prices and return calculations; those are intentionally outside this artifact. **Current artifact path:** G:\My Drive\1amInvesting\Artifacts\WOLF_Wolfspeed_DeepDive.md. **Original snapshot:** versions/originals/WOLF_Wolfspeed_DeepDive.md. Superseded history is retained separately in VERSION_LOG, not in this current-state model.

- **FIN-1** | October 7, 2026 | Wolfspeed conditional facility SEC 8-K. Conditional loan, contribution, inclusive warrant percentage, timing and unresolved final terms.
- **FIN-2** | October 9, 2026 regular session | Archived WOLF quote, USD 30.58, used only for hypothetical contribution pricing; not a live-price refresh or financing announcement.
- **FIN-3** | Post-June primary filings through October 7 | SEC filing review and September proxy/resale documents retained in the run evidence folder. No completed qualifying contribution receipt established; August 31 actual common shares of 52.930569M and the proposed ESPP are distinct from the diluted planning proxy.